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Software engineer and product manager. I build and ship data platforms, AI products, fintech and digital public services across Africa, from the first conversation with users to a system running in production. Selected projects below; walkthroughs and demos available on request.

At a glance

Project	Client / context	Role	Result
National Innovation Portal (RICC)	Federal Ministry of Education, Nigeria	Frontend & UI/UX Lead	Live national platform; offline data collection
NGAO	Field data collection, Kenya	Full-Stack Developer	Rolled out; 3,000+ users
PropFlow	Property management, East Africa	Lead Engineer & Product Lead	In production; users in Kenya, Uganda, Rwanda
Revenue Intelligence Assistant	Nairobi City County	Lead Engineer	Revenue answers in seconds, audited
Cross-Border Investment Platform	Confidential fintech client	Frontend Engineer	Signed off and live in production
Climate & Nature Risk Toolkit	British International Investment	Consultant Developer	Full framework usable by finance staff
MessageHeard	AI video production	Full-Stack Developer	Multi-team platform, transcript to video
Stock Research Assistant	Nairobi Securities Exchange analysts	Lead Engineer	Used by analysts in Kenya
Trademark Registry Upgrade	Federal Ministry of Industry, Trade & Investment, Nigeria	Software Engineer	Upgraded registration system in use
Climate-Stress Analytics	Kenya & Tanzania Bankers Associations	Technical Analyst	Climate risk linked to loan default
Chat Your Finances	Personal finance, Kenya	Lead Developer	AI assistant on WhatsApp, Telegram, web

Featured projects

1. National Innovation Portal (RICC)

Federal Ministry of Education, Nigeria · Frontend & UI/UX Lead · ricc.education.gov.ng

> **Result:** A live national platform, prepared for a national rollout targeting 2,000+ institutional responses, that collects data even where there is no internet.

The problem: Nigeria had no reliable picture of how research moves from universities, institutes and innovators into the market. Data came in through Google Forms, often from places with weak or no internet, and the same innovation was reported again and again. Nobody could see where promising work was getting stuck.

What I did

- Researched why universities, industry, investors and regulators weren't connecting, and turned the findings into user journeys and questionnaires for each stakeholder group.
- Led frontend development of an offline-first web app: field agents complete long questionnaires with no connection, answers are kept on the device and sync automatically once they are back online.
- Built smart questionnaires that hide questions that don't apply, with word limits, ranked choices, state-to-local-government location logic, consent tracking and reference numbers.
- Built the analytics dashboards (readiness scorecards, funding maps, participation by region and a "valley of death" chart of where innovations drop off), an evidence-verification panel for the secretariat, a chatbot that answers with live charts, and terms, consent and privacy controls.
- Ran end-to-end testing on staging before every production release; historical data was cleaned so 62 submissions became 298 distinct innovations with 105 duplicates removed.

Tools: Next.js, React, TypeScript, offline web app (service workers), FastAPI, PostgreSQL

2. NGAO: WhatsApp Data Collection

Last-mile reporting, Kenya · Full-Stack Developer

> **Result:** Rolled out and used by more than 3,000 people.

The problem: Field reporting such as tree planting and seedling sales ran on paper and informal messages: slow, hard to verify and easy to fake.

What I did

- Built the engines and orchestrators that run each conversation: button-led flows in English and Kiswahili that take text, photos, voice notes and location.
- Built answer correction that checks a dictionary first, falls back to AI, asks "Did you mean…?" and learns from every confirmation, so messy answers become clean data.
- Compared each photo's location data with the shared location and flagged mismatches, and kept every submission as a permanent record.
- Connected the backend to the admin dashboards and checked dashboard figures against the raw records and CSV export; a schema builder lets administrators launch new pipelines without new code.

Tools: NestJS, PostgreSQL, React, TypeScript, WhatsApp Business API, Twilio (voice and images), LLMs

3. PropFlow: Property Management Platform

Landlords and managing agents, East Africa · Lead Engineer & Product Lead

> **Result:** In production with users in Kenya, Uganda and Rwanda; partners met in both countries and five CEOs and their CTOs trained in Rwanda.

The problem: Landlords run rent, tenants and repairs across spreadsheets, WhatsApp threads and paper files, so money and documents slip through the cracks.

What I did

- Designed and built, with a partner, one system for the whole tenancy: rent roll and billing, arrears, late fees, deposits and reconciliation, leases generated and signed online, repairs with contractor approvals, online payments and a full audit log.
- Tenants report issues on WhatsApp with photos; emergencies trigger an automatic phone call.
- AI proposes the next step from tenant messages, and every suggestion is checked against business rules before anything happens.
- Shipped production-grade: containerised deployment, background scheduler, nightly backups, error monitoring, automated tests and a written recovery plan. I maintain it.
- Travelled to Uganda and Rwanda to meet partner companies, including the leadership of Keystone Technologies and Unispace, and trained five CEOs, property owners and their CTOs in Rwanda.

Tools: Next.js, TypeScript, Prisma, PostgreSQL, Twilio, WhatsApp, Paystack, Docker, Sentry

4. Revenue Intelligence Assistant

Nairobi City County · Lead Engineer

> **Result:** Revenue staff get correct figures from the county's own data in seconds, in plain words.

The problem: Staff searched monthly spreadsheets in different layouts to answer simple questions about collections, and some money was counted twice.

What I did

- Built a tested import pipeline that reads every report layout into one clean database.
- Added an AI assistant staff can question on WhatsApp or the web; only approved numbers get answers.
- Gave the AI read-only access through a restricted database account, so it can never change records, with automatic switch-over if an AI provider fails.
- Audited answers against the original reports and fixed double-counting and wrong-period results before go-live.

Tools: Next.js, TypeScript, Supabase/PostgreSQL, LLMs, WhatsApp, Vitest

5. Cross-Border Investment Platform

Confidential fintech client · Frontend Engineer

> **Result:** Signed off in user acceptance testing, deployed to production and handed over with source code and documentation.

The problem: People living abroad had no simple, trusted way to invest back home from the apps they already use.

What I did

- Developed and deployed investor onboarding with identity verification (KYC), the marketplace and wallet funding.
- Built for money moving across several countries, within regulatory and foreign-exchange rules, and kept the design working when a key provider changed mid-build.
- Made every amount reconcile exactly, with security planned in advance and an incident playbook ready.

Tools: React, Next.js, Python, PostgreSQL

6. Climate & Nature Risk Toolkit

British International Investment · Consultant Developer

> **Result:** Risk and sustainability teams can work through the whole framework and finish with a complete, shareable record.

The problem: A climate and nature risk framework for financial institutions lived in long PDFs and spreadsheets that finance staff struggled to apply in real decisions.

What I did

- Broke the framework into modules with guided, step-by-step workbenches: maturity assessment, rapid client climate screening, a calculator for the emissions linked to loans and investments, and stress-test calculators.
- Linked records across institutions, clients, projects and portfolios, with CSV import, PDF, Excel and Word exports and admin analytics.

Tools: Next.js, React, TypeScript, Prisma, PostgreSQL

7. MessageHeard: AI Video Production Platform

Media production · Full-Stack Developer

> **Result:** A working multi-team platform that takes a podcast transcript to finished scene images and video.

The problem: A team was producing AI-generated video from podcast transcripts with a single-user prototype that couldn't support several teams at once.

What I did

- Built the backend and the interface, and trained the custom image models.
- Delivered sign-in with roles and invitations and separation between organisations; transcripts that break into timed scenes; an asset library of characters and objects; and AI prompt suggestions per scene.
- Built a generation queue that runs image and video jobs in parallel with live status, retry and cancel, delivered to an 8.5-week plan with weekly client check-ins, CI/CD and a staging environment.

Tools: Next.js, TypeScript, Python, PostgreSQL, Clerk, image and video generation models, LoRA

8. Stock Research Assistant

Nairobi Securities Exchange analysts · Lead Engineer

> **Result:** Used by analysts in Kenya for research and decision support.

The problem: Analysts spend hours gathering prices, Central Bank data and news before they can start analysing a stock.

What I did

- Adapted an open-source multi-agent framework into TypeScript: analyst agents study prices, the economy and news, debate the case for and against, a trader agent makes the call and a risk check tests it against Kenyan inflation.
- Runs automatically every morning before the market opens, with automatic AI-provider failover and a searchable research knowledge base.
- Logs every prediction and scores it later, so the system's accuracy is visible.

Tools: Next.js, TypeScript, Supabase, Vercel AI SDK, GitHub Actions, Redis

9. Trademark Registry Upgrade

Federal Ministry of Industry, Trade and Investment, Nigeria · Software Engineer

> **Result:** An upgraded registration and administration workflow in use by the ministry's administrators.

The problem: The ministry's existing trademark system needed an upgrade to keep up with registrars and applicants.

What I did

- Rebuilt the registration and administration workflows on the upgraded system.
- Kept records consistent from filing to approval and made the admin screens easy for non-technical staff.

10. Climate-Stress Analytics for Banks

Kenya and Tanzania Bankers Associations · Technical Analyst

> **Result:** Climate risk brought into how banks think about credit risk.

The problem: Climate risk wasn't part of how banks judged the chance of loan default.

What I did

- Built exposure analytics linking climate-stress indicators to the probability of default, with NCBA, Equity Bank and Victoria Commercial Bank among the stakeholders.

Tools: Python, data analysis and modelling

11. Chat Your Finances: AI Finance Assistant

Personal finance, Kenya · Lead Developer

> **Result:** An assistant that answers money and tax questions from a person's own statements on WhatsApp, Telegram and the web.

The problem: People struggle to see where their money goes and find filing taxes stressful.

What I did

- Reads M-Pesa, bank and payslip statements, including scanned and password-protected PDFs, using OCR.
- Answers questions from the user's own records using retrieval-based AI; a tax engine reads P9 forms to guide KRA filing; subscription billing through Paystack.

Tools: Next.js, Supabase, OCR, RAG, Paystack

More projects

- **Kenya Climate & Nature Directory:** Power BI dashboards and AI-assisted research and publishing automations, with human review before anything is published.
- **Africa Climate & Nature Data Platform (prototype):** Sector dashboards and a Kenya county-level map of physical and transition climate risk for small businesses. Next.js, Recharts, Leaflet.
- **Process automation: Mediscope and Strathmore Centre for Value Investing:** Mapped how each team really works, designed automations around it and trained the Centre's team to use them.
- **Course evaluation system: Strathmore:** Raised student feedback rates by 75%; the 2026 rebuild adds anonymous one-time links and AI summaries of comments. Django, React, PostgreSQL.
- **Phone-billing system rebuild: Strathmore University ICT:** Rebuilt the university's phone billing as an Odoo module, synced with call records and HR/ERP data. Python, Odoo, Asterisk.
- **Penetration test: Harambee SACCO:** Tested a financial cooperative's systems from an attacker's point of view and set out the fixes.
- **Breach Alert:** Opt-in breach monitoring: owners verify an email by code or a domain by DNS record, emails encrypted at rest, passwords checked without ever being sent in full.
- **Harambee Shop:** Group buying with shared baskets, a discount ladder, supplier-pack splitting, settlement and payouts. Free Pascal (mORMot2), Next.js, PostgreSQL, Redis, Paystack.
- **Tumaini Support & Care:** Healthcare staffing platform with candidate, employer and admin dashboards, verification and audit log. Next.js, Prisma, Playwright.
- **Strathmore faculty tools:** Faculty workload management app (Django, React) and admissions data pipelines (Python, Selenium).
- **Event-streaming pipeline:** Apache Kafka on a three-broker cluster with Python services in Docker and PostgreSQL.
- **Data science (WorldQuant University):** ARMA air-quality forecasting, GARCH volatility forecasting served via an API, bankruptcy prediction, earthquake damage classification with a bias audit, customer segmentation, A/B testing with ETL, regression price models.